

SAP USR02 User Analysis

AI-Assisted ABAP Development

Case Study using Watsonx.ai

Business Requirement

- Obtain list of users with validity dates
- Track failed logon attempts
- Capture last logon date and time
- Focus on active users from SAP table USR02

AI-Based Solution

Watsonx.ai (llama-3-3-70b-instruct) was used to generate and enhance ABAP code.

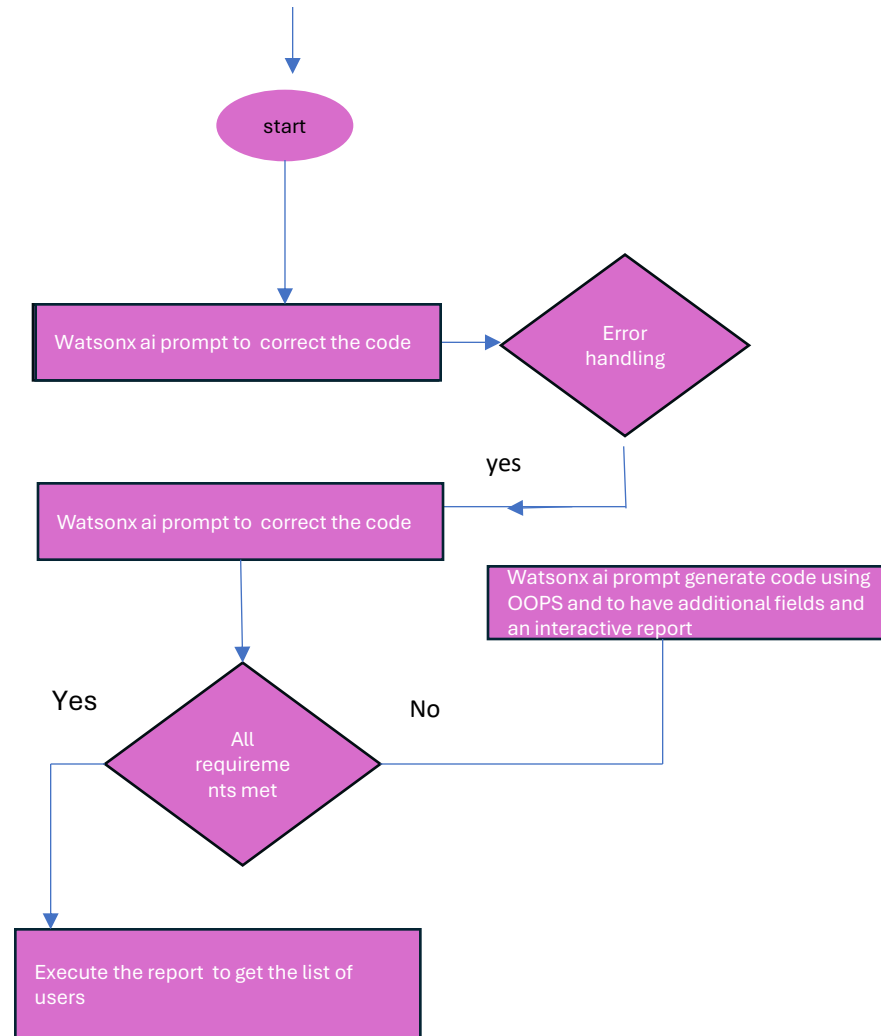
- Generated object-oriented ABAP code
Resolved syntax errors
Added user group and additional fields
Created interactive report with SU01 navigation on double-click

Benefits & Productivity Gains

- Reduced manual development effort. Estimated development time reduced from 5-6 hours to 30-40 minutes. Improved requirement understanding and stakeholder communication. Supported timely project execution

Implementation and Governance

- AI-generated solution refined and executed in SAP ABAP editor
Technical specifications prepared using IBM Consulting Advantage
All AI-generated outputs reviewed before use
Risks addressed: incorrect responses, bias, and data privacy concerns
Human judgment applied throughout



AI coding in ABAP using Watson.ai

Output from Table

Table: USR02

Displayed Fields: 44 of 44

Fixed Columns: 2

List Width: 1023

	MANDT	BNAME	BCODE	GLTGV	GLTGB	USTYP	CLASS	LOCNT	UFLAG	ACCNT	ANAME	ERDA
☐	103	DEHR138_07	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	13.1
☐	103	DEHR138_08	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	13.1
☐	103	DEHR140_03	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	13.1
☐	103	DEHR140_04	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	13.1
☐	103	DEHR648_01	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	13.1
☐	103	DEHR648_02	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	13.1
☐	103	02CCOCCCADE0	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	12.0
☐	103	02CCOCCCADEA	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	12.0
☐	103	02CCOCCCADED	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	12.0
☐	103	02CCOCCCADEN	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	26.1
☐	103	02CCOCCIC1P0	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	25.0
☐	103	02CCOCCOCDE0	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	12.0
☐	103	02CCOCCOCDEA	0000000000000000	00.00.0000	00.00.0000	A	GCEUR_DEYY	0	0		JOBRUN	12.0

The Problem statement

 IBM watsonx

Upgrade

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Projects / N Sathya's sandbox / Prompt Lab



☐ AI guardrails off 

Unsaved  

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npts  

Chat | Structured | Freeform | AI

Model: llama-3-3-70b-instruct 

  TXT  

 You: 02:14 PM

Give me a ABAP report to display the list of users from USR02 in SAP system including fields last logon Date , time and account group

ing transcript summary
arize the discussion from a
ng transcript.

Create a report in ABAP editor

[illegible]



Title: List of users

Original language: EN

Created: NBNSA1 30.06.2026

Last Changed:

Status: New(Revised)

Attributes

Type: Executable program

Status: Unclassified

Authorization Group:

Application:

LDB name:

Selection screen:

ABAP Language Version: Standard ABAP

- REPORT zuser_list.

- TYPES: BEGIN OF ty_user,

- bname TYPE usr02-bname, "Username
- gltgb TYPE usr02-gltgb, "Last Logon Date
- gltgs TYPE usr02-gltgs, "Last Logon Time
- class TYPE usr02-class, "Account Group
- END OF ty_user.

- DATA: lt_users TYPE TABLE OF ty_user.

- SELECT bname gltgb gltgs class FROM usr02 INTO TABLE lt_users.

- WRITE: / 'Username', 20 'Last Logon Date', 40 'Last Logon Time', 60 'Account Group'.

- ULINE.

- LOOP AT lt_users INTO DATA(ls_user).

- WRITE: / ls_user-bname, 20 ls_user-gltgb, 40 ls_user-gltgs, 60 ls_user-class.

- ENDLOOP.

Errors in the code while performing syntax check

 ABAP Editor: Change Report Z_LISTOFUSERS



More ▾



Report: inactive (revised)

```
2 report z_listofusers.
3
4 TYPES: BEGIN OF ty_user,
5         bname TYPE usr02-bname, "Username
6         gltgb TYPE usr02-gltgb, "Last Logon Date
7         gltgs TYPE usr02-gltgs, "Last Logon Time
8         class TYPE usr02-class, "Account Group
9     END OF ty user.
```

1 errors for Program Z_LISTOFUSERS



Type C...	Description
	Program Z_LISTOFUSERS
	No component exists with the name "GLTGS", but the following components with similar names exist: "GLTGB" and "GLTGV".

Giving the errors to Watsonx.ai

The screenshot shows the IBM Watsonx AI interface. The top navigation bar includes "Projects / N Sathya's sandbox / Prompt Lab", a checkmark icon, a toggle for "AI guardrails off", and a status "Unsaved". Below this, a sidebar on the left contains a list of prompts, including "Generating transcript summary". The main chat area has tabs for "Chat", "Structured", and "Freeform", with "AI" selected. The model is set to "llama-3-3-70b-instruct". A message from the user at 02:29 PM states: "Error in the above code 'No component exists with the name GLTGS but the following components with similar names exist GLTG8 and GLTGV'".

This screenshot shows the same IBM Watsonx AI interface, but with a response from the AI. The top navigation bar is identical. The sidebar on the left now includes an additional prompt: "Scenario classification". The main chat area shows a section titled "Corrected Code". The text explains: "To fix the error, you need to use the correct field names. The GLTGV field represents the last logon date and time in the USR02 table. You can use the following code to display the list of users with last logon date and time:". Below this, a code block contains the following SQL code:

```
REPORT zuser_list.  
  
TYPES: BEGIN OF ty_user,  
        bname TYPE usr02-bname, "Username  
        gltgb TYPE usr02-gltgb, "Last Logon Date  
        gltgv TYPE usr02-gltgv, "Last Logon Time  
        class TYPE usr02-class, "Account Group  
END OF ty_user.
```

No syntax error in the code

The screenshot displays the SAP ABAP Editor interface for Report Z_LISTOFUSERS. The menu bar includes Program, Edit, Goto, Utilities, Environment, System, and Help. The title bar shows the report name and its status as 'inactive (revised)'. The code editor contains the following ABAP code:

```
1> report z_listofusers.
2>
3>
4> TYPES: BEGIN OF ty_user,
5>         bname TYPE usr02-bname, "Username
6>         gltgb TYPE usr02-gltgb, "Last Logon Date
7>         gltgv TYPE usr02-gltgv, "Last Logon Time
8>         class TYPE usr02-class, "Account Group
9>       END OF ty_user.
10>
11> DATA: lt_users TYPE TABLE OF ty_user.
12>
13> SELECT bname gltgb gltgv class FROM usr02 INTO TABLE lt_users.
14>
15> WRITE: / 'Username', 20 'Last Logon Date', 40 'Last Logon Time', 60 'Account Group'.
16> ULINE.
17>
18> LOOP AT lt_users INTO DATA(ls_user).
19>   WRITE: / ls_user-bname, 20 ls_user-gltgb DD/MM/YYYY, 40 ls_user-gltgv, 60 ls_user-class.
20> ENDLOOP.
21>
```

The status bar at the bottom indicates the scope is \LOOP, the language is ABAP, and the cursor is at line 19, column 50. A message box at the bottom left states: "No syntax errors found in Report Z_LISTOFUSERS". Buttons for "Save" and "Cancel" are located at the bottom right.

To enhance the code using Watsonx.ai

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Projects / N Sathya's sandbox / Prompt Lab

AI guardrails off Unsaved

Chat Structured Freeform AI Model: llama-3-3-70b-instruct

Chat

You: 02:42 PM

Can you prepare an ALV report using classes

watsonx 02:42 PM

ALV Report using Classes

Here's an example of how to create an ALV report using classes in ABAP:

```
REPORT zalv_report.

CLASS lcl_alv DEFINITION.
PUBLIC SECTION.
CLASS METHODS:
```

< SAP ABAP Editor: Change Report Z_LISTOFUSERS

Report: Z_LISTOFUSERS inactive (revised)

```
32 lr_columns->get_column( 'ACCNT' )->set_short_text( 'Account' ).
33 lr_columns->get_column( 'ACCNT' )->set_medium_text( 'Account' ).
34 lr_columns->get_column( 'ACCNT' )->set_long_text( 'Account' ).
35 lr_columns->get_column( 'TRDAT' )->set_short_text( 'Transaction Date' ).
36 lr_columns->get_column( 'TRDAT' )->set_medium_text( 'Transaction Date' ).
37 lr_columns->get_column( 'TRDAT' )->set_long_text( 'Transaction Date' ).
38
39
```

2 errors for Program Z_LISTOFUSERS

68 2 13 0

Type	Lo...	Line	Description
●		37	Program Z_LISTOFUSERS
			The literal "Transaction Date" is not type-compatible with the formal parameter "VALUE".

Error given as input in Watsonx in

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AI guardrails off

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Unsaved

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Chat

Structured

Freeform

AI

Model: llama-3-3-70b-instruct

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N.

You: 03:07 PM

error in the above code ' the literal transaction date is not compatible with formal parameter value

🤖

watsonx 03:07 PM

Error Analysis

The error message 'the literal "Transaction Date" is not compatible with formal parameter value' suggests that the `set_short_text` , `set_medium_text` , and `set_long_text` methods are expecting a different type of parameter.

Code Review

The `set_short_text` , `set_medium_text` , and `set_long_text` methods are expecting a parameter of type

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Chat

Structured

Freeform

AI

Model: llama-3-3-70b-instruct

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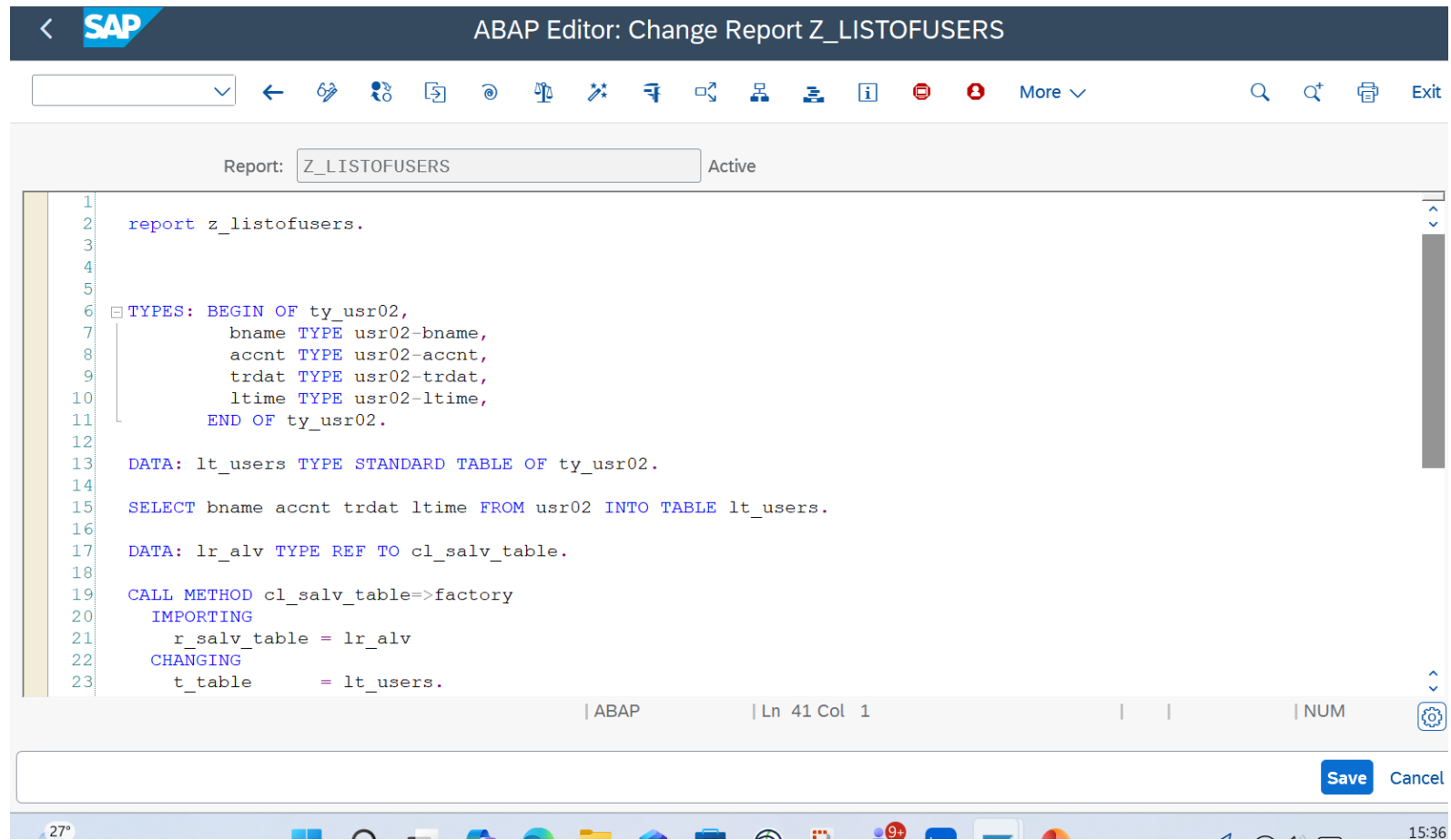
TXT

</>

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Extend the code by adding the field ACCNT.

No syntax errors



The screenshot displays the SAP ABAP Editor interface for the report Z_LISTOFUSERS. The code is as follows:

```
1 report z_listofusers.
2
3
4
5
6 TYPES: BEGIN OF ty_usr02,
7         bname TYPE usr02-bname,
8         acctn TYPE usr02-accnt,
9         trdat TYPE usr02-trdat,
10        ltime TYPE usr02-ltime,
11      END OF ty_usr02.
12
13 DATA: lt_users TYPE STANDARD TABLE OF ty_usr02.
14
15 SELECT bname acctn trdat ltime FROM usr02 INTO TABLE lt_users.
16
17 DATA: lr_alv TYPE REF TO cl_salv_table.
18
19 CALL METHOD cl_salv_table=>factory
20     IMPORTING
21       r_salv_table = lr_alv
22     CHANGING
23       t_table      = lt_users.
```

The editor interface includes a top bar with the SAP logo and the title "ABAP Editor: Change Report Z_LISTOFUSERS". Below this is a toolbar with various icons for editing and navigation. The main area shows the code with line numbers on the left. The status bar at the bottom indicates the current position as "Ln 41 Col 1" and includes "Save" and "Cancel" buttons.

Output form the report

☰ ListEditGotoSettingsSystemHelp				
<  L				
More				
User	Account number	Logon	Logon	
NBKONANKGO		03.07.2026	12:32:03	
FRSIMONAN			00:00:00	
NBNALISW		03.07.2026	12:56:07	
NBGUPTASA2		03.07.2026	12:56:28	
NBKONKARHA		03.07.2026	12:57:07	
FROHLMANJU		03.07.2026	12:58:55	
NBRPA1		03.07.2026	13:06:46	
BRJUNIORDI1		03.07.2026	13:14:59	
OSS_97565		03.07.2026	13:15:55	
NBTPA		03.07.2026	13:17:52	
TEST_SUT_03			00:00:00	
NBVISHALVI		03.07.2026	13:19:28	
NBMETWALLO		03.07.2026	13:30:12	
NBADAPASH1		03.07.2026	13:34:42	
PNUTHET		03.07.2026	13:38:10	
XSPULUSARA		03.07.2026	13:46:17	